Innovating Financial Inclusion: The Role of Digital Technologies and Financial Literacy in Emerging Economies

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Abstract

Globally, 1.7 billion adults lack access to formal financial services, primarily in South Asia and Sub-Saharan Africa. Current financial inclusion strategies, in line with the United Nations 2030 Agenda for Sustainable Development Goals—specifically targeting 700 million people living on less than $1.90 per day—aim to empower these populations. While financial inclusion isn't explicitly listed as a goal, it is now recognized as a vital catalyst for achieving the SDGs, serving as a powerful tool to eliminate poverty (Chibba, 2009; Klapper, El-Zoghbi, & Hess, 2016; Lyons & Kass-Hanna, 2020a). This study investigates the impact of digital technology and financial literacy on financial inclusion in 56 countries across South Asia and Sub-Saharan Africa from 2009 to 2019. The study employs IV-TOBIT regression to handle the limited financial inclusion variable range and mitigate endogeneity concerns. Recognizing IV-TOBIT's limitations, GMM techniques are also used to address issues like unobserved heterogeneity and reverse causality in estimation. Findings reveal that digital technology, measured by ICT indicators and a composite indicator of digitalization, significantly and positively affects financial inclusion. A robustness check, utilizing a composite indicator of financial inclusion, further confirms the positive association. The study emphasizes the need for increased investments in financial, technological infrastructures, and human capital sectors, as financial literacy plays a crucial role in promoting stability and inclusive finance in emerging economies.

Keywords: financial inclusion, digital technology, digital literacy, poverty eradication, financial literacy, United Nations Sustainable Development Goals (SDG), universal financial access, Sub-Saharan Africa, South Asia.

Academic Integration:

This capstone project explores the intersection of financial inclusion, financial literacy and digital technology in achieving the United Nations Sustainable Development Goals (SDGs), particularly in the contexts of Sub-Saharan Africa and South Asia. The research draws upon a wide range of academic literature and theoretical frameworks to inform its analysis and findings.

One key area of academic integration is the conceptualization of financial inclusion and its multidimensional nature. Building on the works of scholars such as Demirgüç-Kunt and Klapper (2012) and Bateman (2015), this project adopts a holistic approach to financial inclusion, encompassing not only access to formal financial services but also issues of affordability, quality, and usage. Additionally, the research engages with theories of technology adoption and diffusion, drawing upon models proposed by Rogers (2003) and Venkatesh et al. (2003). By integrating these theories into the analysis, the project seeks to understand the factors influencing the adoption of

digital financial services in low-income communities and the implications for financial inclusion efforts.

Introduction

In the domain of financial and economic development, financial inclusion stands as a pivotal concept, as articulated by Raichoudhury (2020). It encompasses intentional measures aimed at fostering access and utilization of financial services. Broadly defined, financial inclusion refers to the extent to which financial services are tailored to encompass excluded segments, such as low-income earners or vulnerable populations within an economy, as delineated by Sinclair (2013). The significance of financial inclusion is underscored by its multifaceted advantages. By facilitating avenues for savings, asset accumulation, consumption management through borrowing, business initiation or expansion, and investments in education and health, it provides individuals with the tools to create a financial safety net and withstand economic shocks. The widespread acknowledgment of access to financial services as a crucial step in reducing poverty and inequality emphasizes its positive impact on the lives and economic prospects of people globally.

Financial inclusion serves as a pivotal policy instrument endorsed by various developmental entities to attain Universal Financial Access (UFA) and align with Sustainable Development Goals (SDG) (OECD, 2013; Robert et al., 2005). This alignment extends to the majority of UN SDGs and the AU Agenda 2063. Particularly pertinent for emerging economies, notably in Sub-Saharan Africa and South Asia, the imperative of achieving financial inclusivity for societies previously excluded from financial services has led to the development of comprehensive financial inclusion strategies (Zins and Weill, 2016; Demirgüç-Kunt et al., 2018). The UN 2030 Agenda, which introduced the SDGs, does not explicitly mention financial inclusion; however, its significance has become increasingly apparent as a catalyst for SDGs, especially in the context of poverty elimination (Chibba, 2009; Klapper, El-Zoghbi, & Hess, 2016; Lyons & Kass-Hanna, 2020a). Globally recognized as a critical catalyst for poverty alleviation and reduced inequality, access to financial services yields benefits such as improved savings, entrepreneurship, and resilience against financial shocks (Lyons et al., 2020). In response to this recognition, concerted global efforts aim for universal financial inclusion, reflected in the substantial increase in the percentage of adults holding financial accounts, rising from 62.0% in 2014 to 69.0% in 2018 (Demirgüç-Kunt et al., 2018).

Financial inclusion plays a pivotal role in extending services to unbanked segments, facilitating business expansion, and contributing to socioeconomic development. Academic attention has shifted towards the intricate relationship between digital technologies and financial inclusion, aiming to understand whether the proliferation of digital technology serves as a facilitator of financial development or introduces potential risks to system stability (Asongu et al., 2021; Nchofoung & Asongu, 2022a). Despite advancements in mobile money transfers, challenges arising from low financial literacy rates and associated technological costs necessitate a critical examination of the role of digital technologies in enhancing levels of inclusive finance. Furthermore, digital technologies have been argued to be an asset for the inclusion of the financial system and for achieving sustainable development (Asongu et al., 2021; Nchofoung and Asongu,

2022 a). This study departs from this perspective with a focus on non-developed economies such as Sub-Saharan Africa and South Asia.

Firstly, the fastest-growing countries today in the world are emerging economies, as indicated by the GDP growth forecasts from the IMF's October 2023 World Economic Outlook. Unsurprisingly, many of these countries are located in South Asia and Sub-Saharan Africa—two of the world's fastest-growing regions. The growth in the number of account owners, mobile phone users, and internet penetration in these countries signifies financial and economic development. This motivates the exploration of the extent to which access to digital technologies enhances financial literacy and promotes inclusive finance in emerging economies such as Sub-Saharan Africa (SSA) and South Asia (SA).This is a crucial inquiry considering the rapid economic growth of these countries, which is surpassing developed nations.

Digital technologies can ensure the accessibility of financial services by providing them at affordable prices to the less privileged. Evans (2018) documented that developed countries have higher access to finance than developing countries, primarily due to advanced technology, which plays a significant role in reducing financial exclusion. This is because digitalization contributes to reaching out to the poor on time and provides access to less expensive information and better connectivity through mobile banking, internet banking, and mobile automated teller machine services. As I delve into these inquiries, this study seeks to inspire a deeper comprehension of the transformative capacity of digital technologies in reshaping financial landscapes and fostering inclusive finance in emerging economies. This motivation underscores the importance of investigating the extent to which access to digital technologies enhances financial literacy and promotes inclusive finance, particularly in regions like Sub-Saharan Africa (SSA) and South Asia (SA).

The idea of a multifaceted approach to addressing ICT is appropriate because it allows for a comprehensive examination of various indicators, providing a more holistic understanding of the research topic. By incorporating all commonly used ICT indicators and introducing a composite indicator focusing on financial inclusion metrics, the study aims to pioneer an investigation into the impact of digital technologies on financial inclusion in South Asia and Sub-Saharan Africa. The inclusion of indicators such as the number of individuals using the Internet, fixed broadband subscribers, fixed phone subscribers, and mobile cellular phone subscribers reflects a thorough exploration of various dimensions of digital technologies. Furthermore, the research employs the IV-TOBIT regression method to address the constrained range of financial inclusion variables and mitigate potential endogeneity concerns. Recognizing the limitations of IV-TOBIT regression, the study incorporates Generalized Method of Moments (GMM) techniques. GMM is particularly employed to handle unobserved heterogeneity by considering time-invariant omitted variables and addressing concerns related to reverse causality. The primary GMM technique is implemented to overcome these specific estimation challenges. Therefore, the research methodology is apt for its objective of empirically verifying the effect of digital technology on financial inclusion in South Asia and Sub-Saharan Africa, given its comprehensive use of ICT indicators and the application of suitable regression techniques to address methodological challenges.

The finding in this study suggests that increased access to digital technologies fosters literacy and financial inclusivity. This implies that the rise in financial inclusion can be attributed to

advancements in digital technology in these regions. To arrive at this conclusion, the remainder of the paper is organized into five sections. Section 2 delves into the relevant theoretical literature and empirical findings on the nexus between digital technologies and financial inclusion. Section 3 outlines the data and methodology employed in the study. In Section 4, the results are presented and discussed in detail. Section 5 provides a robustness analysis to strengthen the findings, and finally, Section 6 offers the conclusion drawn from the research findings.

Literature Review

Studies on digital technologies and financial inclusion have not been well-explored. Some research on digital technologies and development indicators has predominantly focused on the economic and environmental domains, neglecting how technological growth affects financial inclusion and deepening. In attempts to investigate the determinants of financial inclusion, Evans (2018) found that digital technologies significantly influence and increase the likelihood of individuals participating in the financial market. The authors further assert that ICT enhances financial inclusion in both the informal and formal sectors. Regarding the influence of digital banking, Siddik et al. (2017) discovered that digitalization through internet banking has increased financial inclusion and boosted the performance of banks in Bangladesh. In their examination of the financial inclusion-growth nexus, Lenka and Barik (2018) documented that digitalization through mobile phone and internet services penetration positively affects the degree of inclusion in financial services, contributing to inclusive growth. The authors also revealed a linear causal relationship between the degree of digital technologies and expanded financial inclusion. In the same empirical findings, the authors posit a significant positive role of information and communication technologies in financial infrastructure, boosting the rate of financial inclusion. Triki and Faye (2013) researched the nexus between digital technology, especially the impact of mobile technology on financial inclusion in Africa, and concluded that mobile technologies facilitate financial services transactions in the financial sector, enabling many unbanked individuals to be included in the financial sector and to have access to financially affordable services.

The most interesting research relevant to this has been conducted by J.C. Koulamdoum et al. to investigate the intricate relationship between digital technology and financial inclusion in 43 Sub-Saharan African countries spanning the period from 2004 to 2019. Utilizing the Generalized Method of Moment (GMM) to account for double causality and country heterogeneity, along with IV-Tobit to address the constrained range in dependent variables, the study uncovers consistently positive and statistically significant effects of digital technology on financial inclusion across various levels. Employing diverse ICT indicators, encompassing fixed and mobile telephone subscription rates, fixed broadband, internet users, and a composite measure of digitalization, the research illustrates a positive association between the increasing prevalence of digital technologies and heightened rates of financial inclusion in Sub-Saharan Africa. A rigorous robustness check, employing a composite indicator of financial inclusion, further validates the constructive influence of digital technology on financial inclusion.

This study contributes to existing relevant literature by comprehensively incorporating all pertinent ICT indicators, including fixed telephone, fixed broadband, internet users, and mobile cellular subscriptions, to examine their impact on financial inclusion in South Asia and Sub-

Saharan African countries. The strategic deployment of digital technology within the financial sector emerges as a crucial instrument for raising awareness, underscoring the imperative for heightened investments in financial and technological infrastructure, along with human capital development. This emphasis aims to foster financial literacy and stability in these regions. Noteworthy is the study's expansion of the empirical scope beyond conventional banking sectors, shedding light on the often-overlooked dimensions of mobile and digital banking, which have received limited attention in prior research centered on African countries.

Data Collections and Empirical Methodology

The research relies on secondary data obtained from the World Bank development indicator, encompassing a dataset comprising 56 countries situated in both Sub-Saharan Africa and South Asia. The temporal scope of the data spans from the year 2009 to 2019, allowing for a comprehensive examination of trends, patterns, and dynamics over this decade-long period. This extensive dataset serves as a valuable resource, providing a robust foundation for the empirical analysis conducted in the study. The inclusion of countries from these regions enhances the geographical diversity of the data, enabling a more nuanced understanding of the digital technology and financial inclusion nexus in a broad and varied context. By leveraging this rich dataset, the study aims to derive meaningful insights into the intricate relationship between digital technologies and financial inclusion across these diverse nations.

I. Dependent Variable

The focal dependent variable in this study is financial inclusion, operationalized through various metrics such as bank accounts per 1000 adults, bank branches per 100,000 adults, and ATMs subscribers per 100,000 adults. Additionally, the study employs a composite indicator of financial inclusion, created via Principal Component Analysis (PCA). To maintain consistency with established methodologies, all indicators are normalized to a standardized range between 0 and 1, aligning with the procedural framework proposed by Ngouhouo and Nchofoung (2022). The selection of these specific indicators is grounded in their widespread adoption within the existing literature, as acknowledged by scholars such as Sha'ban, Girardone, and Sarkisyan (2020), Evans (2016), Turvey and Xiong (2017), and Shen et al. (2018). Each chosen indicator serves a distinct purpose in capturing facets of financial inclusion. The count of bank accounts elucidates the growing participation of individuals in the financial realm, reflecting the demand-side dynamics of financial inclusion. Meanwhile, the density of bank branches offers insights into the geographical dispersion of financial services, concurrently influencing the concentration ratio within the banking sector. Finally, the metrics pertaining to deposits and credit within the banking sector signify the impact of financial service provision on broader economic activities.

The composite indicator obtained through PCA analysis, along with its factors, can be specified as:

$$FIndex_{it} = Z_{j1}X_1 + Z_{j2}X_2 + Z_{j3}X_3 + Z_{j4}X_4 + \cdots + Z_{jp}X_p \qquad (1)$$

Findex represents the financial inclusion index, where Z_j signifies the weight of the parameter of the factor score, and X represents the values of the respective components. Here, 'p' represents

the number of variables used in the analysis. The computation of this index using PCA analysis is inspired by the work of Anarfo (2018).

When constructing the composite financial inclusion index, we considered two dimensions, each comprising three factors. The first dimension involves supply-side factors, including ATMs per 100,000 adults, bank branches per 100,000 adults, and commercial bank branches per 100,000 adults. Conversely, the second dimension encompasses demand-side factors such as deposits, domestic credit, and bank loans. Hence, the Financial Inclusion Index (FINDEX) is derived from six variables. The index can be formulated as follows:

FINDEX = f (NUMOFATM, BBRANCH, BACCOUNT, DEPOSIT, DOMCREDIT, BLOAN) (2)

II. Independent Variable/Variable of Interest:

The primary independent variable of interest is digital technology, which is assessed through four indicators: the number of individuals using the Internet, the number of fixed broadband subscribers, the number of fixed phone subscribers, and mobile cellular phone subscribers. Additionally, a composite indicator encompasses all these mentioned indicators. These indicators have been employed to yield valuable policy implications for various facets of digital technologies (Sha'ban et al., 2020; Evans, 2016, 2018), anticipating a positive impact on financial inclusion, as technology is widely perceived as a catalyst for financial development and inclusive finance. The numbers of internet users, fixed broadband, fixed telephone, and mobile cellular subscribers all exhibit positive relationships with financial inclusion indicators, including the number of bank accounts per 100,000 population, the number of branches, and mobile money subscribers. Correlation analysis underscores the positive impact of ICT growth on the rate of financial inclusion.

III. Control Variables:

Controlled variables encompass diverse factors that could potentially influence financial inclusion in Sub-Saharan Africa and South Asia, incorporating elements such as gross domestic product, the rate of domestic savings, government expenditures, foreign direct investments, and trade. Widely recognized in the literature, these variables serve as fundamental determinants of financial inclusion. The proposition of foreign direct investment as a determinant stems from the research of Odugbesan et al. (2020), which suggests a positive effect on financial inclusion. The anticipation of negative impacts on financial inclusion due to changes in consumer prices, incorporating inflation as a determinant, aligns with the findings of Sha'ban et al. (2020). The utilization of trade as an indicator of financial inclusion finds its grounding in the study conducted by Lenka and Sharma (2017). Finally, the consideration of government final consumption as a determinant draws inspiration from the research of Makina and Walle (2019) and Turegano and Herrero (2018), while the inclusion of domestic savings as a determinant is inspired by the works of Rillo and Miyamoto (2016).

Table 1 provides an overview of the key statistics pertaining to the variables under consideration. The table illustrates that the dependent variables exhibit a range between 0 and 1, displaying diverse standard deviation values.

Table 1: Descriptive Statistics of SSA and SA

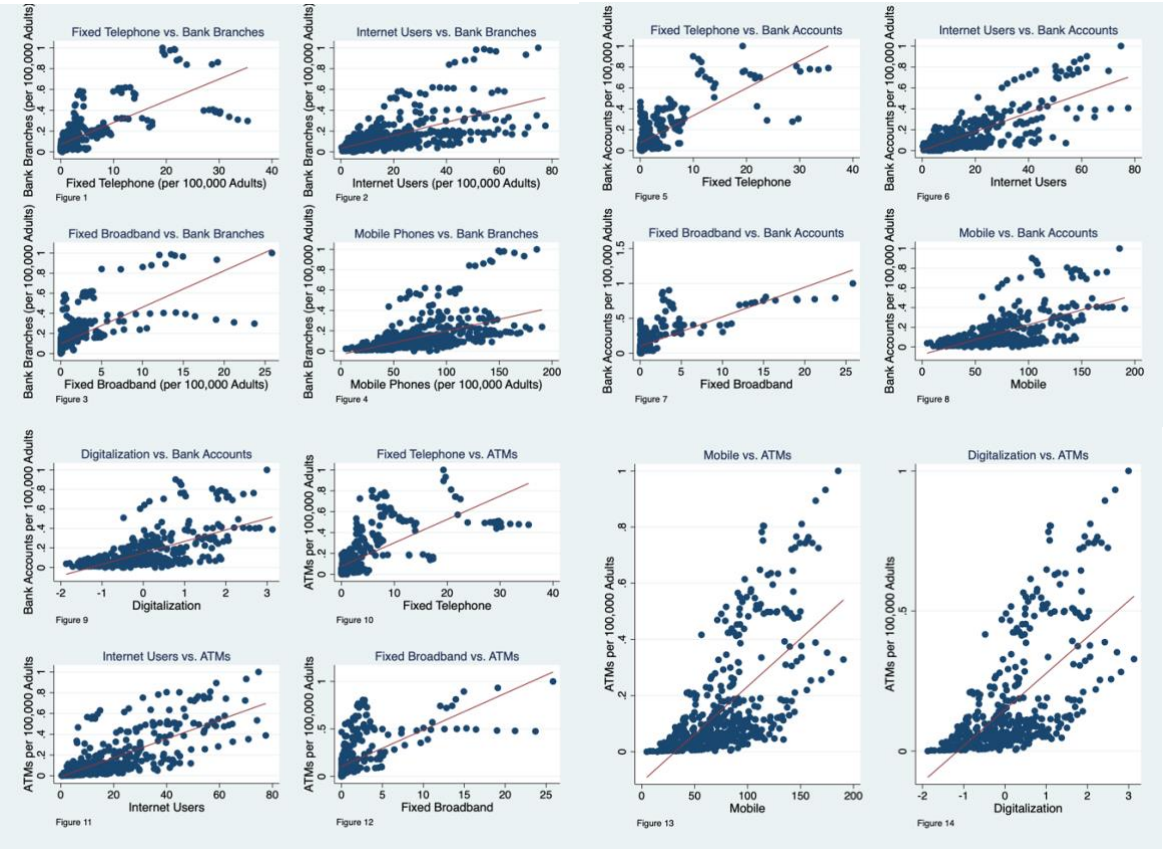
Variable	Obs	Mean	Std. Dev.	Min	Max
Bank account	417	.145	.183	0	1
ATMS	513	.148	.191	0	1
Bank branches	561	.137	.162	0	1
Financial inclusion Index	256	. 216	.24	0	1
Fixed telephones	605	2.924	5.482	0	35.386
Internet Users	590	16.359	16.091	.26	77.47
Fixed broadband	575	1.187	2.989	0	25.844
Mobile	611	72.637	36.951	4.576	190.525
digitalization	552	21.57643	13.89282	0.2342211	74.12599
Foreign direct investment	597	4.562	8.868	-11.197	103.337
GDP per capita	599	4.161	4.87	-46.082	21.452
Consumption	530	14.457	6.964	3.588	43.484
Saving	543	15.404	17.708	-53.552	68.283
Inflation	599	7.309	15.61	-26.7	225.395
Trade	544	69.738	34.9	15.282	222.082

Note: The summary statistics presented below provide an overview of the dependent variables, the variable of interest, and the control variables for countries in Sub-Saharan Africa (SSA) and South Asia (SA).

The descriptive statistics provided in Table 1 offer valuable insights into various indicators related to financial inclusion, communication technology adoption, and economic variables across Sub-Saharan Africa (SSA) and South Asia (SA). Among these indicators, mobile phone subscriptions stand out prominently, with an average of 72.637 subscriptions per 100,000 adults. This suggests widespread access to mobile communication technology, potentially facilitating greater connectivity and access to financial services among individuals in these regions. In contrast, both fixed broadband and fixed telephone subscriptions exhibit comparatively lower levels of development, with mean values of 1.187 and 2.924, respectively, per 100,000 adults. This disparity underscores the dominance of mobile communication technology over traditional fixed-line services. Furthermore, the data reveals that digitalization, with an average value of 21.57643, is a significant factor shaping technological advancement and, consequently, financial inclusion. The mean value of internet users (16.359 per 100,000 adults) highlights the importance of internet connectivity in enabling access to a wide range of financial services and information. However, it's essential to note the limited variability observed in financial inclusion variables such as bank accounts, ATMs, and bank branches, with mean values ranging from 0.137 to 0.216.

The scatter plots presented in Figures 1 to 14 offer an initial glimpse into the relationship between digital technology and financial accessibility over time. Through a visual examination of these plots, we explore how advancements in digital technology have influenced access to bank accounts, bank branches, and ATMs, thereby shaping the landscape of the financial sector. Each scatter plot depicts the relationship between various ICT indicators, such as fixed telephone subscriptions, fixed broadband, internet users, mobile phone subscriptions, and the composite indicator of digitalization, with indicators of financial inclusion, including the number of bank accounts, bank branches, and ATMs. Notably, the scatter plots reveal distinct patterns and trends across different ICT indicators.

Upon closer inspection, it becomes evident that mobile phone subscriptions have emerged as a dominant force in driving financial inclusion, as evidenced by their prominence in Figures 4, 8, and 13. Conversely, indicators such as fixed broadband (depicted in Figures 3, 7, and 12) and fixed telephone subscriptions (illustrated in Figures 1, 5, and 10) exhibit comparatively slower rates of development. In fact, fixed telephone subscriptions have witnessed a decline on a global scale since the advent of mobile phones, underscoring the transformative impact of mobile technology on communication and financial access. These observations provide valuable insights into the evolving relationship between digital technology and financial inclusion. The increasing prevalence of mobile phone subscriptions highlights the pivotal role of mobile technology in expanding access to financial services, particularly in regions where traditional banking infrastructure may be limited. Conversely, the stagnation or decline of fixed broadband and fixed telephone subscriptions underscores the need for continued efforts to bridge the digital divide and ensure equitable access to digital financial services for all segments of society. Additionally, the numbers on the x-axis reflect the rapid expansion of each ICT indicator over the years, further emphasizing the dynamic nature of technological progress and its implications for financial inclusion.



Empirical Methodology

The study employs the IV-TOBIT regression method to address the restricted range of financial inclusion variables and mitigate potential endogeneity concerns (Nchofoung, Asongu, Njamen Kengdo, & Achuo, 2022b). The linearization of the dependent variable is necessary for

the application of instrumental Tobit regression. This method requires the dependent variable to have a limited range. To meet this condition, the dependent variable is linearized using the min-max technique. Linearizing the dependent variable ensures that it falls within a restricted range, making it suitable for analysis using instrumental Tobit regression. This approach allows for the estimation of relationships between the dependent variable and other variables of interest while accounting for censoring or limitations in the observed data.

The instrumental variables used for IV-Tobit are the first lag of ICT indicators such as internet users, fixed broadband, and fixed and mobile cellular phone users. Upon the regression with the first lag in as Table 2, the result shows that these indicators can be used as proxied for digital technology to determine the ICT influence on financial literacy and inclusion. In addition, GMM techniques are applied to overcome the limitations of IV-TOBIT regression in handling issues related to unobserved heterogeneity, encompassing accounting for time-invariant omitted variables and addressing concerns of reverse causality. The primary GMM technique is implemented to tackle these estimation challenges, covering the period from 2009 to 2019 and focusing on 56 Sub-Saharan Africa and South Asia countries. This selection aligns with the critical condition for adopting the GMM strategy, which necessitates a greater number of individuals than the time series (Nchofoung, Asongu, Njamen Kengdo, & Achuo, 2022b; Nchofoung and Asongu, 2022b).

Furthermore, the study fulfills the conditions requisite for adopting GMM in panel data analysis. The system GMM technique employed in the study is succinctly summarized with the following equations in levels (3) and in the first difference (4):

(3) $FI_{it} = \beta_0 + \beta_1 FI_{i(t-1)} + \beta_2 Tech_{it} + \sum_{h=1}^{k} \delta_h Z_{hi(t-\tau)} + \mu_i + \gamma_t + \varepsilon_{it}$

(4) $FI_{it} - FI_{i(t-\tau)} = \beta_1\left(FI_{i(t-\tau)} - FI_{i(t-2\tau)}\right) + \beta_2\left(Tech_{it} - \beta_2 Tech_{i(t-\tau)}\right) + \sum_{h=1}^{k} \delta_h (Z_{hi(t-\tau)} - Z_{hi(t-\tau)}) + (\gamma_t - \gamma_{(t-\tau)}) + (\varepsilon_{it} - \varepsilon_{i(t-\tau)})$

In the model, the variable FI denotes financial inclusion, quantified by metrics such as the count of bank accounts, bank branches, and ATM users. The variable Tech represents digital technologies, proxied by ICT indicators encompassing internet users, fixed broadband, and fixed and mobile cellular phone users. The vector Z incorporates control variables, including Foreign Direct Investments, GDP, savings, government spending, trade, and inflation. In the panel analysis, μ_i represents the country-specific effect, γ_t signifies the time-specific constant, ε_{it} denotes the error term, and τ represents the lagging coefficient.

To address key challenges inherent in the GMM framework, such as identification, simultaneity, and exclusion restrictions, all explanatory variables are treated as exogenous. Time-fixed effects are introduced as instruments in both the difference and level equations, employing the forward orthogonal deviation to mitigate instrument proliferation. Additionally, an instrumental Tobit regression is implemented. To satisfy the conditions necessary for applying IV-Tobit, the limited range of the dependent variable is addressed by linearizing it through the min-max technique (Ngouhouo & Nchofoung, 2022). Internal instruments are utilized in the IV-Tobit computation, involving regressing the suspected endogenous explanatory variables with their first-

period lags and using the resulting residuals as instruments, following established literature (Nchofoung, Achuo, & Asongu, 2021).

Results and Discussions

As a result of my study, I employed IV-Tobit regression to address the censored dependence variables, which range between 0 and 1. This approach also aimed to mitigate concerns regarding endogeneity, ensuring that endogenous variables are not correlated with the error term. Endogenous variables, such as the presence of conflict in these regions, accumulation of debts, and corrupt financial institutions, could potentially affect the outcomes but are not included or controlled for in the analysis. To address this, instrumental variables (IVs) were utilized for the IV-Tobit regression. These IVs were derived from digital technology indicators, such as internet usage, fixed broadband subscriptions, and fixed and mobile phone ownership. Each IV corresponds to different regression, facilitating a comprehensive analysis of the impact of digital technology on financial inclusion.

However, IV-Tobit regression alone does not address the concern of unobserved heterogeneity, which includes time-invariant omitted variables and the issue of reverse causality. To overcome this limitation, I employed the Generalized Method of Moments (GMM) estimation technique. GMM allows for the control of time-invariant omitted variables by taking the first lag of the independent variables into account. My study spans the period from 2009 to 2019 and focuses on 56 countries in South Asia and Sub-Saharan Africa. This fulfills a key condition for adopting the GMM strategy, which requires a sufficient number of individuals relative to the time series (Nchofoung, Asongu, Njamen Kengdo, & Achuo, 2022b; Nchofoung and Asongu, 2022b). The analysis began with the baseline results of the IV-Tobit regression, which addressed the limited range of the study variables. These baseline results are presented in three different tables, each representing a different indicator of financial inclusion. The first table examines the effects of ICT indicators on the number of bank accounts per 100,000 people. The second table investigates how ICT indicators influence the number of bank branches per 100,000 people in the regions. Finally, Table 4 presents the IV-Tobit estimates on how the number of automated teller machine (ATM) users per 100,000 population is influenced by digital technology indicators, such as internet usage, fixed broadband subscriptions, and fixed and mobile phone subscriptions per 100,000 people in the regions.

The IV-TOBIT results presented (see Table 2) demonstrate a statistically significant positive effect of digital technologies on the number of bank accounts. These results are further elucidated and interpreted in Table 5, which presents GMM estimates. Table 3 reveals a positive effect of digital technology indicators on the number of bank branches per 100,000 adults in South Asia and Sub-Saharan Africa, indicating that the presence of digital technologies correlates with an increase in the number of branches. The causal relationship between ICT and bank branches is further explored in Table 6. Findings from Table 4 indicate a positive effect of digital technologies on the number of automated teller machines (ATMs) per 100,000 adults in Sub-Saharan Africa and South Asia. These results are subjected to further analysis in a more robust system of GMM estimates presented in Table 7 (also see Table 8).

Analysis from Table 5 (Table 5) reveals that an increase in the number of individuals using the

internet (0.0590***) is associated with a rise in the number of bank accounts per 100,000 adults in both Sub-Saharan Africa and South Asia. This suggests that internet penetration in these regions promotes inclusivity in the financial sector, facilitated by the availability of internet services that enable activities such as E-banking. Moreover, the number of bank accounts also increases with an uptick in fixed phone subscriptions (0.044***), which exhibits a positive and statistically significant effect. Similarly, an increase in mobile phone subscriptions (0.023***) correlates with a higher number of bank accounts. Additionally, the number of fixed broadband subscriptions demonstrates a positive and statistically significant effect on the number of bank accounts in Sub-Saharan Africa and South Asia (0.080***). Utilizing the number of bank accounts per 100,000 adults as a proxy for financial inclusion leads to the conclusion that digital technology plays a role in enhancing financial inclusion in the banking sector of these regions. This measure has been adopted by other researchers such as Anarfo (2018).

Table 6 reveals a positive relationship between the number of bank branches and all ICT indicators. Specifically, the number of telephone subscriptions (0.0617***), and mobile cellular(0.0417***) exhibits a statistically significant effect on the number of bank branches, suggesting that bank branches increase with an increase in phone subscriptions. Additionally, indicators of internet penetration, such as the number of individuals (0.0954***) and fixed broadband (0.0475***), positively influence the number of bank branches per 100,000 adults in Sub-Saharan Africa. This signifies that internet penetration has led to the inclusion of many individuals in the financial sector, as indicated by the increasing number of individuals using the internet and the number of fixed broadband subscribers per 100,000 adults. Consequently, digital technology contributes to a higher rate of financial inclusion in Sub-Saharan Africa and South Asia. Furthermore, the digitalization composite indicator shows a positive and statistically significant effect on financial inclusion measured by the number of bank branches (0.208***).

In Table 7, where financial inclusion is measured by ATM users per 100,000 adults, the results demonstrate a positive effect of internet subscribers (0.0142***), fixed broadband (0.0185***), fixed telephone users (0.0181***), mobile cellular subscription (0.0541***), and the ICT composite indicator (0.0191***) on the number of automated teller machines per 100,000 adults. These findings indicate that the number of ATM users increases with an increase in digital technologies, aligning with the conclusions of Sha'ban et al. (2020), who highlighted the benefits of digital technology in driving financial sector development and increasing the rate of financial inclusion. The positive effects of digitalization on financial inclusion are consistent with the findings of other studies by Sha'ban et al. (2020), Turvey and Xiong (2017), and Evans (2016).

Table 2: The effect of digital technology on financial inclusion in SA and SSA (IV-Tobit)

	Dependent variable:	Bank account			
Variables	(1)	(2)	(3)	(4)	(5)
Fixed telephone	0.0258***				
	(0.00146)				
Internet users		0.00965***			
		(0.000539)			
Mobile phone			0.00277***		
			(0.000301)		
Fixed broadband				0.0454***	
				(0.00287)	

Digitalization					0.108***
					(0.0122)
Foreign Direct	0.00402	-0.00251**	-0.00277	-0.00284***	-0.00198***
Investment	(0.00135)	(0.00135)	(0.00162)	(0.00179)	(0.00222)
GDP per capita	-0.00125	-0.000185	-0.00320	-0.00302	-0.00720**
	(0.00161)	(0.00162)	(0.00203)	(0.00214)	(0.00271)
Consumption	-0.000774	-0.00220	-0.00445***	0.00289	-0.00501***
	(0.00141)	(0.00140)	(0.00179)	(0.00157)	(0.00188)
Savings	0.00121***	-0.000633	-0.00130	0.000772	-0.00140
	(0.000525)	(0.000528)	(0.000693)	(0.000565)	(0.000733)
Inflation	0.000686	0.000228	0.000296	0.000559	0.000138
	(0.000376)	(0.000372)	(0.000476)	(0.000395)	(0.000487)
Trade	0.00598**	0.00831***	0.00194**	0.00361***	0.00206***
	(0.000335)	(0.000329)	(0.000396)	(0.000378)	(0.000437)
Constant	0.0423	-0.00869	-0.0865**	0.0403	0.141***
	(0.0217)	(0.0214)	(0.0278)	(0.0245)	(0.0346)
Observation	327	324	331	306	299

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Table 3: The effect of digital technology on financial inclusion in SA and SSA (IV-Tobit)

	Dependent variable:	Bank branch			
Variables	(1)	(2)	(3)	(4)	(5)
Fixed telephone	0.0177***				
	(0.000954)				
Internet users		0.00543***			
		(0.000409)			
Mobile phone			0.00171***		
			(0.000187)		
Fixed broadband				0.0340***	
				(0.00218)	
Digitalization					0.0628***
					(0.00738)
Foreign Direct	0.0828***	0.0258***	0.0158***	0.0268***	0.0838***
Investment	(0.0853)	(0.000986)	(0.00103)	(0.00105)	(0.00124)
GDP per capita	-0.0885	0.00128	-0.000273	-0.000305	-0.00199
	(0.0116)	(0.00140)	(0.00145)	(0.00151)	(0.00191)
Consumption	-0.00279	-0.00305**	-0.00361**	0.00115	-0.00385**
	(0.00101)	(0.00115)	(0.00126)	(0.00115)	(0.00131)
Savings	0.00119***	0.000356	-0.000223	0.000841*	-0.000509
	(0.000364)	(0.000418)	(0.000457)	(0.000405)	(0.000481)
Inflation	0.000106	-0.000431	-0.000388	-0.000213	-0.000475
	(0.000296)	(0.000339)	(0.000370)	(0.000324)	(0.000382)
Trade	0.00110***	0.00192***	0.00243***	0.00102***	0.00243***
	(0.000239)	(0.000259)	(0.000272)	(0.000274)	(0.000291)
Constant	-0.0102***	-0.0380***	-0.0941***	-0.0381***	-0.0421***
	(0.0150)	(0.0177)	(0.0204)	(0.0173)	(0.0219)
Observation	437	427	441	414	401

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Table 4: The effect of digital technology on financial inclusion in SA and SSA (IV-Tobit)

	Dependent variable:	ATMs			
Variables	(1)	(2)	(3)	(4)	(5)
Fixed telephone	0.0220***				
	(0.00125)				
Internet users		0.00906***			
		(0.000459)			
Mobile phone			0.00296***		
			(0.000225)		
Fixed broadband				0.0415***	
				(0.00287)	
Digitalization					0.116***
					(0.00898)
Foreign Direct	-0.00231***	-0.000373	-0.00288*	-0.00232	-0.00246
Investment	(0.00111)	(0.00108)	(0.00120)	(0.00138)	(0.00146)
GDP per capita	-0.00113	-0.000254	-0.00314	-0.00399*	-0.00729***
	(0.00149)	(0.00151)	((0.00167)	(0.00195)	(0.00221)
Consumption	0.0110***	0.00727***	0.00613***	0.0130***	0.00633***
	(0.00132)	(0.00126)	(0.00147)	(0.00151)	(0.00154)
Savings	0.00281***	0.00110*	0.000775	0.00235***	0.000959
	(0.000476)	(0.000458)	(0.000534)	(0.000532)	(0.000566)
Inflation	0.000566	-0.000320	-0.000257	-0.000141	-0.000430
	(0.000380)	(0.000367)	(0.000428)	(0.000418)	(0.000444)
Trade	-0.000555	0.000115	0.000899**	-0.000629	0.000843*
	(0.000315)	(0.000288)	(0.000321)	(0.000364)	(0.000346)
Constant	-0.0662***	-0.112***	-0.205***	-0.0462*	0.0337
	(0.0197)	(0.0196)	(0.0240)	(0.0227)	(0.0261)
Observation	407	397	411	384	371

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Foreign direct investments (FDI) demonstrate a positive and statistically significant effect on the number of bank branches (0.0569***), while appearing to have a negative significant effect on some equations of bank accounts. This disparity suggests that financial development impacts various indicators of financial inclusion differently across regions. The positive effect of FDI on bank branches is attributed to investments in the financial sector, which increase the number of banks and their branches, particularly in areas previously underserved by financial institutions. This finding aligns with that of Odugbesan et al. (2020).

However, the findings regarding the effect of GDP and inflation on financial inclusion are mixed and vary depending on different measures. A negative effect of GDP per capita on the number of bank accounts (-0.0518***), bank branches (-0.0424***), and ATM users (-0.0384***) implies that GDP growth in Sub-Saharan Africa and South Asia coincides with a decrease in the number of individuals engaging in financial services. This could be due to the possibility that high rates of inclusive finance lead to an increase in lending rates, resulting in reduced demand for funds and potential long-term negative effects on businesses. Conversely, a negative effect of inflation on financial inclusion is understandable, as rising prices can impede financial activities. These results are consistent with the findings of Mushtaq and Bruneau (2019) and Lenka and Sharma (2017).

Trade exhibits a positive and statistically significant effect on the financial inclusion index (0.00856***), indicating that financial inclusion in Sub-Saharan Africa increases with an uptick in trade in goods and services. This finding aligns with that of Lenka and Sharma (2017).

Conversely, government final consumption demonstrates a negative effect on financial inclusion (-0.00529), suggesting that spending patterns may not adequately consider the majority of the population. This finding is consistent with the results of Makina and Walle (2019). Additionally, domestic savings exert a positive influence on the rate of financial inclusion in Sub-Saharan Africa (0.00282***), confirming the findings of Rillo and Miyamoto (2016).

Table 5: The effect of digital technology on financial inclusion in SA and SSA(GMM)

	Dependent variable:	Bank account			
Variables	(1)	(2)	(3)	(4)	(5)
L.Bank account	0.975***	0.990***	1.005***	0.921***	0.997***
	(0.0272)	(0.0433)	(0.0300)	(0.0743)	(0.0290)
Fixed telephone	0.044***				
	(2.161)				
Internet users		0.0590***			
		(1.115)			
Mobile phone			0.023***		
			(0.289)		
Fixed broadband				0.080***	
				(9.544)	
Digitalization					0.370***
					(11.94)
Foreign Direct	0.235**	0.119	0.0955	-0.122	0.739
Investment	(0.399)	(0.685)	(0.404)	(0.944)	(0.524)
GDP per capita	-0.0410***	-0.0653***	-0.079***	-0.0143***	-0.0518***
	(0.00752)	(0.00639)	(0.0774)	(0.0114)	(0.0285)
Consumption	-0.083***	-0.020***	-0.145***	-0.251	-0.434***
	(0.138)	(0.507)	(0.594)	(0.878)	(0.628)
Savings	-0.266***	-0.143	-0.407	-0.291	-0.325
	(0.338)	(0.463)	(0.484)	(0.398)	(0.397)
Inflation	0.271***	0.285**	0.263**	0.277*	0.240*
	(0.0992)	(0.0864)	(0.0903)	(0.118)	(0.105)
Trade	0.415***	0.899*	0.786*	0.376	0.940*
	(0.326)	(0.402)	(0.374)	(0.382)	(0.403)
Constant	1.84***	-1.42***	-1.78***	3.76***	7.376***
	(0.67)	(0.66)	(0.11)	(0.05)	(0.21)
Observation	323	322	326	310	306
Prob > AR1	0.030	0.023	0.035	0.020	0.053
Prob > AR2	0.149	0.130	0.134	0.119	0.165
Prob > Hansen	0.232	0.234	0.235	0.315	0.335

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Table 6: The effect of digital technology on financial inclusion in SA and SSA(GMM)

	Dependent variable:	Bank branch			
Variables	(1)	(2)	(3)	(4)	(5)
L.Bank branch	0.974***	1.012***	1.009***	1.009***	1.010***
	(0.0207)	(0.0108)	(0.00854)	(0.0157)	(0.0103)
Fixed telephone	0.0617***				
	(0.0405)				
Internet users		0.0954***			
		(0.00537)			
Mobile phone			0.0417***		
			(0.00223)		
Fixed broadband				0.475***	
				(0.0484)	
Digitalization					0.208***
					(0.101)

Foreign Direct	0.00412	0.0125***	0.0569***	0.0439*	0.0934**
Investment	(0.00775)	(0.00826)	(0.00768)	(0.00757)	(0.00843)
GDP per capita	-0.0424***	-0.0689**	-0.00863	-0.0170*	-0.0252***
	(0.0105)	(0.0134)	(0.0117)	(0.0217)	(0.0273)
Consumption	-0.0346*	-0.0294	-0.0235	-0.0255*	-0.0172***
	(0.0158)	(0.0172)	(0.0170)	(0.0140)	(0.0150)
Savings	-0.0706**	-0.00592	-0.00508	-0.00454*	-0.00388***
	(0.00740)	(0.00659)	(0.00714)	(0.00656)	(0.00616)
Inflation	-0.0525***	-0.0548***	-0.0494***	-0.0433**	-0.0438**
	(0.00136)	(0.00144)	(0.00131)	(0.00130)	(0.00141)
Trade	0.0671***	0.0873***	0.0609***	0.0676***	0.0758***
	(0.00484)	(0.00552)	(0.00469)	((0.00484)	(0.00523)
Constant	0.374***	0.241***	0.471***	0.139***	-0.118***
	(0.263)	(0.271)	(0.267)	(0.241)	0.338)
Observation	438	431	441	424	414
Prob > AR1	0.021	0.021	0.021	0.022	0.021
Prob > AR2	0.100	0.100	0.101	0.102	0.093
Prob > Hansen	0.241	0.265	0.534	0.280	0.283

Standard errors in parentheses
***p<0.01, **p<0.05, *p<0.1

Table 7: The effect of digital technology on financial inclusion in SA and SSA(GMM)

	Dependent variable:	ATMs			
Variables	(1)	(2)	(3)	(4)	(5)
L.ATMs	0.987***	1.041***	1.033***	0.984***	1.031***
	(0.0208)	(0.0244)	(0.0180)	(0.0339)	(0.0226)
Fixed telephone	0.0181***				
	(0.0549)				
Internet users		0.0142***			
		(0.0189)			
Mobile phone			0.0541***		
			(0.00603)		
Fixed broadband				0.0185***	
				(0.162)	
Digitalization					0.0191***
					(0.284)
Foreign Direct	0.0109	0.0414**	0.076***	0.0430***	0.0945***
Investment	(0.0127)	(0.0122)	(0.0140)	(0.0203)	(0.0160)
GDP per capita	-0.0231*	0.0194	-0.0140	-0.0384***	-0.0292***
	(0.0181)	(0.0174)	(0.0198)	(0.0332)	(0.0351)
Consumption	-0.0135***	-0.0438***	-0.0291	0.0606***	-0.0275***
	(0.0254)	(0.0268)	(0.0280)	(0.0239)	(0.0255)
Savings	0.0185**	0.0144***	0.0127	0.0151***	0.0151**
	(0.00846)	(0.00910)	(0.00869)	(0.0105)	(0.00906)
Inflation	-0.00493	-0.00396**	-0.00355	-0.00569	-0.00457
	(0.00313)	(0.00338)	(0.00310)	(0.00364)	(0.00338)
Trade	0.00688	0.0161*	0.0161**	0.0111	0.0142
	(0.00767)	(0.00811)	(0.00851)	(0.00898)	(0.00883)
Constant	-0.0429***	-0.0839***	-0.0345***	-0.0818***	-0.243***
	(0.558)	(0.423)	(0.558)	(0.528)	(0.556)
Observation	400	393	403	386	376
Prob > AR1	0.079	0.070	0.073	0.074	0.074
Prob > AR2	0.958	0.985	0.998	0.994	0.991
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes
Prob > Hansen	0.210	0.347	0.368	0.0627	0.277

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Robustness analysis

To ensure the robustness of the research findings, I conducted a comprehensive analysis by computing a composite indicator using Principal Component Analysis (PCA). This composite indicator incorporated various financial inclusion metrics commonly found in the literature, including the number of bank accounts (Table 5), ATMs (Table 7), and bank branches (Table 6) per 100,000 adults, as well as the amount borrowed from and deposited with commercial banks per 100,000 adults. Upon computing this composite index of financial inclusion, I found that it yielded consistent results with those obtained from the individual indicators. Specifically, all indicators of digital technologies exhibited positive effects on financial inclusion in the regions, affirming the significant impact of digital technology on inclusive finance initiatives. The methodology of computing this composite index with the inclusive finance indicators draws inspiration from the work of Anarfo (2018), who similarly employed PCA to construct a composite indicator of financial inclusion. This approach enhances the robustness of the analysis by capturing the multifaceted nature of financial inclusion and synthesizing various relevant metrics into a single comprehensive measure.

When assessing the progress of individual countries in embracing digital technologies, it becomes apparent that there is relatively little variation observed across regions. While South Asia appears to demonstrate a slightly more pronounced advancement in this regard, the differences, upon closer examination, are relatively subtle and do not represent a significant departure from the trends observed in Sub-Saharan Africa. A deeper dive into the summary statistics of digital technology adoption within Sub-Saharan Africa and South Asia, excluding any outliers, sheds further light on this comparison. Within Sub-Saharan Africa, while there are pockets of notable progress and innovation in certain countries, the overall landscape still reflects a significant digital divide. Challenges such as limited infrastructure, access barriers, and socioeconomic disparities continue to impede widespread adoption and integration of digital technologies across the region. South Asia exhibits a somewhat more accelerated pace of digital transformation, driven by factors such as rapid urbanization, expanding middle-class demographics, and government initiatives aimed at fostering digital inclusion. However, similar to Sub-Saharan Africa, disparities persist within individual countries, with rural areas often lagging behind urban centers in terms of digital infrastructure and access to technology-driven opportunities. While the summary statistics(see Appendix 1) may hint at disparities between Sub-Saharan Africa and South Asia, it's crucial to recognize the nuanced complexities underlying these trends. Understanding the unique challenges and opportunities within each region is essential for devising targeted interventions and policies aimed at bridging the digital divide and ensuring equitable access to the benefits of digital innovation.

Table 8: The effect of digital technology on financial inclusion in SA and SSA(GMM)

	Dependent variable:	Financial Inclusion	Index		
Variables	(1)	(2)	(3)	(4)	(5)
L.Financial Inclusion	0.851***	0.863***	0.864***	0.828***	0.873***
Index	(0.0744)	(0.0724)	(0.0761)	(0.0683)	(0.0650)
Fixed telephone	0.0210***				
	(0.0101)				
Internet users		0.0558***			
		(0.00245)			
Mobile phone			0.0341***		
			(0.00201)		

Fixed broadband				0.0424***	
				(0.0178)	
Digitalization					0.101***
					(0.0608)
For Direct Investment	-0.00143***	-0.000180	-0.00152*	-0.000112***	-0.00101***
	(0.00157)	(0.00154)	(0.00200)	(0.00156)	(0.00191)
GDP per capita	-0.0584***	-0.00260***	-0.00453**	-0.00833***	-0.0134***
	(0.0137)	(0.0141)	(0.0143)	(0.0178)	(0.0188)
Consumption	0.00267	0.00135	-0.00529	0.00910	-0.00275***
	(0.00544)	(0.00570)	(0.00600)	(0.00674)	(0.00592)
Savings	0.00193***	0.00133***	-0.00114***	0.00282***	0.000216***
	(0.00195)	(0.00206)	(0.00249)	(0.00195)	(0.00202)
Inflation	0.0103***	0.0966***	0.0707***	0.0105***	0.0614***
	(0.000853)	(0.000756)	(0.000815)	(0.000913)	(0.000902)
Trade	0.00856***	0.00802***	0.00126***	0.00100***	0.00112***
	(0.00106)	(0.00666)	(0.00631)	(0.00128)	(0.00648)
Constant	-0.0795***	-0.177***	-0.219***	-0.115***	0.0236***
	(0.151)	(0.177)	(0.179)	(0.157)	(0.171)
Observation	205	205	206	196	194
Prob > AR1	0.038	0.051	0.042	0.040	0.038
Prob > AR2	0.090	0.087	0.087	0.110	0.100
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes
Prob > Hansen	0.450	0.368	0.453	0.274	0.347

Standard errors in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Conclusions

In conclusion, this research paper utilized IV-Tobit and GMM methodologies to conduct empirical analysis on the impact of digital technology on financial inclusion across 56 countries in South Asia and Sub-Saharan Africa from 2009 to 2019. The study focused on four ICT indicators, including internet usage, fixed telephone subscriptions, mobile cellular subscriptions, and fixed broadband, along with a composite indicator of digitalization encompassing these metrics. Financial inclusion was assessed using variables such as the number of bank accounts per 100,000 adults, bank branches, ATM users per 100,000 adults, domestic credit to private businesses, deposit amounts, and borrowed money, aggregated into a composite indicator via PCA analysis. Additionally, GDP, inflation, and FDI were considered as determinants of financial inclusion based on literature review.

IV-Tobit and GMM methodologies were employed to address various estimation challenges. IV-Tobit addressed the limited range of financial inclusion variables after normalization, while GMM tackled potential endogeneity and reverse causality issues in the results. The findings indicate significant effects of digital technology on financial inclusion in the regions, underscoring the crucial role of ICT in promoting inclusive finance, particularly in regions where access to financial services is limited for a considerable portion of the population. These results align with prior studies by Sha'ban et al. (2020) and Turvey and Xiong (2017).

However, when examining the regions individually, the differences in the impact of digital technology on financial inclusion appear to be less pronounced. This suggests a more uniform effect across the regions. Therefore, this research contributes to the understanding of the relationship between digital technology and financial inclusion, emphasizing the importance of ICT in driving inclusive finance initiatives in underserved regions.

Appendix 1:

Table 9: Descriptive Statistics for SSA Data

Variable	Obs	Mean	Std. Dev.	Min	Max
Bank Account	373	332.048	449.737	0	2424.75
Automatic Teller Machine	431	13.48	18.148	.1	89.99
Bank Branch	473	6.992	9.361	.04	55.07
Internet Users	507	15.48	15.527	.26	74.756
Fixed Telephone	517	2.757	5.646	0	35.386
Fixed Broadband	488	1.011	3.066	0	25.844
Mobile Phone	523	70.273	36.401	4.576	185.559
Foreign Direct Investment	509	4.958	9.386	-11.192	103.337
GDP per capita	511	3.88	4.998	-46.082	21.452
Consumption	441	14.911	7.107	3.588	43.482
Savings	394	17.783	11.88	-12.881	57.85
Inflation	495	8.351	23.59	-16.86	380
Trade	455	70.727	34.023	15.282	222.082

Table 10: Descriptive Statistics for SA Data

Variable	Obs	Mean	Std. Dev.	Min	Max
Bank Account	44	511.085	343.425	87.49	1196.39
Automatic Teller Machine	82	12.899	10.444	.45	48.09
Bank Branch	88	10.905	4.801	1.87	19.31
Internet Users	83	21.725	18.389	1.97	77.47
Fixed Telephone	88	3.905	4.297	.019	17.314
Fixed Broadband	87	2.173	2.289	.004	10.539
Mobile Phone	88	86.688	37.285	20.712	190.525
Foreign Direct Investment	88	2.118	3.345	-.639	16.783
GDP per capita	88	5.627	3.425	-8.158	21.391
Consumption	61	10.778	4.51	5.039	21.901
Savings	61	31.236	9.371	11.375	48.687
Inflation	88	6.028	3.557	-6.811	13.648
Trade	61	50.905	23.599	24.702	115.924

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